

Week 9: Networking Fundamentals

Date: May 4, 2026

Study Time: 5-8 hours total

GOAL

Understand your network's IP structure completely

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" by Brian Ward
 - Chapter 9: Understanding Your Network
 - Focus on: TCP/IP basics, network interfaces, routing

Video Learning

- ☐ Professor Messer - Search: "TCP/IP Fundamentals"
 - <https://youtube.com/@professormesser>
- ☐ LearnLinuxTV - Search: "IP Addressing Linux"
 - <https://youtube.com/@LearnLinuxTV>

Online Practice

- ☐ Subnetting Practice - <https://subnettingpractice.com>
 - Practice calculating subnets
 - Understand CIDR notation (/24, /16, etc.)

Concepts to Understand

- ☐ IPv4 addressing (192.168.x.x)
- ☐ Public vs Private IP addresses
- ☐ CIDR notation (what does /24 mean?)
- ☐ Network interfaces (eth0, wlan0, tailscale0)
- ☐ Routing tables
- ☐ DNS basics
- ☐ Common ports (22=SSH, 80=HTTP, 443=HTTPS)
- ☐ TCP vs UDP

LAB WORK (3-4 hours)

Understand Your Current Network

☐ View all network interfaces

bash

```
ip addr show
# OR the legacy command:
ifconfig

# What you'll see:
# - lo (loopback - 127.0.0.1)
# - eth0/enp0s3 (Ethernet)
# - wlan0/wlp2s0 (WiFi)
# - tailscale0 (your Tailscale interface)

# Note each interface's IP address
```

☐ View specific interface

bash

```
ip addr show eth0
ip addr show tailscale0
```

Document ALL Device IP Addresses

Create a network map. For EACH device, document:

☐ Ubuntu Laptop

bash

```
# On laptop:
ip addr show

# Record:
# - Local IP: _____
# - Tailscale IP: _____
# - Subnet mask: _____
# - Gateway: _____
```

☐ Windows Desktop

bash

```
# On Windows (PowerShell or CMD):  
ipconfig /all  
  
# Record same info
```

☐ Fedora Server (Tier 2)

bash

```
# SSH into Fedora:  
ssh fedora  
ip addr show  
  
# Record IPs
```

☐ Raspberry Pi 4 (Tier 4)

bash

```
ssh pi4  
ip addr show
```

☐ Raspberry Pi Zero (Tier 3)

bash

```
ssh pi-zero  
ip addr show
```

Understand Routing

☐ View routing table

bash

```
ip route show
# OR legacy:
route -n

# You'll see:
# - default via X.X.X.X = your gateway/router
# - 192.168.1.0/24 dev eth0 = local network
# - 100.64.0.0/10 dev tailscale0 = Tailscale network
```

☐ Trace route to internet

bash

```
traceroute google.com
# OR on some systems:
traceroute -I google.com

# Shows each hop your traffic takes
# First hop = your router
# Last hop = google's server
```

☐ Trace route within Tailscale

bash

```
traceroute your-pi-tailscale-ip
# Should be direct (1 hop)
```

Test Network Connectivity

☐ Ping your router/gateway

bash

```
ping 192.168.1.1
# OR whatever your gateway is

# Should show response times
# Ctrl+C to stop
```

☐ Ping each device in your infrastructure

bash

```
# From laptop, ping each device:
ping fedora-tailscale-ip
ping pi4-tailscale-ip
ping pi-zero-tailscale-ip

# Verify all respond
```

☐ Ping internet

bash

```
ping 8.8.8.8 # Google DNS
ping google.com # Tests DNS too
```

Check Listening Ports and Services

☐ View all listening ports

bash

```
# Modern way:
ss -tulpn

# Legacy way:
netstat -tulpn

# Flags:
# -t = TCP
# -u = UDP
# -l = listening
# -p = show process
# -n = show numbers (not names)
```

☐ Find what's using a specific port

bash

```
# What's on port 22 (SSH)?
ss -tulpn | grep :22

# What's on port 80 (HTTP)?
ss -tulpn | grep :80
```

☐ Check on all your devices

```
bash
```

```
# On each device:
```

```
ssh device
```

```
ss -tulpn
```

```
# Document what services are running where
```

Draw Your Tailscale Network Topology

Create a diagram showing:

```
Internet
```

```
|
```

```
├─ Tier 1 (Local Network - 192.168.1.x)
```

```
|
```

```
├─ Ubuntu Laptop (192.168.1.10, Tailscale: 100.x.x.1)
```

```
|
```

```
├─ Windows Desktop (192.168.1.11, Tailscale: 100.x.x.2)
```

```
|
```

```
└─ iPhone (192.168.1.12, Tailscale: 100.x.x.3)
```

```
|
```

```
├─ Tier 2
```

```
|
```

```
├─ Fedora Server (192.168.1.100, Tailscale: 100.x.x.10)
```

```
|
```

```
├─ Gitea (port 3000)
```

```
|
```

```
└─ Centralized Logging (port 514)
```

```
|
```

```
├─ Tier 3
```

```
|
```

```
├─ Pi Zero (Bastion) (192.168.1.101, Tailscale: 100.x.x.20)
```

```
|
```

```
├─ Tier 4
```

```
|
```

```
├─ Pi 4 (Automation/Exit Node) (192.168.1.102, Tailscale: 100.x.x.30)
```

```
└─ Exit Node (routes internet traffic)
```

Use paper, draw.io, or any tool you like.

Subnetting Practice

☐ Understand /24 notation

```
192.168.1.0/24 means:
```

```
- Network: 192.168.1.0
```

```
- Usable IPs: 192.168.1.1 - 192.168.1.254
```

```
- Broadcast: 192.168.1.255
```

```
- Subnet mask: 255.255.255.0
```

```
- Total hosts: 254
```

☐ Practice on subnettingpractice.com

- Do 10 practice problems
 - Focus on /24, /16, /8 first
 - Understand how to calculate ranges
-

FLASHCARD COMMANDS

Create flashcards for these commands:

1. `ip addr`
2. `ip addr show`
3. `ip route`
4. `ip route show`
5. `ping`
6. `traceroute`
7. `netstat`
8. `netstat -tulpn`
9. `ss`
10. `ss -tulpn`
11. `ifconfig` (legacy)
12. `route -n` (legacy)

Concepts for Flashcards:

- What is a /24 network?
 - What is a default gateway?
 - Difference between TCP and UDP?
 - What port does SSH use?
 - What is 127.0.0.1?
 - What is a private IP address range?
-

EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: [https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant Sections:

- 1.3 Given a scenario, configure and verify network connection parameters
 - 1.5 Given a scenario, use the appropriate networking tools
 - 2.1 Given a scenario, analyze system properties and remediate accordingly
-



NOTES & REFLECTIONS

My Network Map: (Sketch or describe your topology here)

Device IP Addresses:

- Ubuntu Laptop: Local _____ Tailscale _____
- Windows Desktop: Local _____ Tailscale _____
- Fedora Server: Local _____ Tailscale _____
- Pi 4: Local _____ Tailscale _____
- Pi Zero: Local _____ Tailscale _____

Services Discovered: (What ports are open on each device?)

Subnetting Practice Results: (How many did you get right?)

Questions to Research:

Completed: ☐



QUICK REFERENCE

Common Private IP Ranges:

- 10.0.0.0/8 (10.0.0.0 - 10.255.255.255)
- 172.16.0.0/12 (172.16.0.0 - 172.31.255.255)
- 192.168.0.0/16 (192.168.0.0 - 192.168.255.255)

Tailscale IPs:

- 100.64.0.0/10 (CGNAT range)

Common Ports:

- 22: SSH
- 80: HTTP
- 443: HTTPS
- 3000: Gitea (default)
- 514: Syslog
- 53: DNS

CIDR Cheat Sheet:

- /32 = 1 IP (255.255.255.255)
- /24 = 254 IPs (255.255.255.0)
- /16 = 65,534 IPs (255.255.0.0)
- /8 = 16,777,214 IPs (255.0.0.0)

Troubleshooting Network Issues:

```
bash
```

```
# 1. Check interface is up
```

```
ip link show
```

```
# 2. Check IP address assigned
```

```
ip addr show
```

```
# 3. Check can reach gateway
```

```
ping your-gateway
```

```
# 4. Check can reach internet
```

```
ping 8.8.8.8
```

```
# 5. Check DNS works
```

```
ping google.com
```

```
# 6. Check routing
```

```
ip route show
```

Previous Week: Systemd Custom Services

Next Week: Firewall Rules